

Learning Disocclusion Reliability: When and Where a Generative Prior Helps Single-View Reconstruction

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Abstract

Single-view 3D reconstruction systems fail unevenly: some disoccluded regions can be reconstructed by a feed-forward model, while others require generative completion. We study the reliability prediction problem: before invoking a slow generative teacher, can we predict whether it will help — and where? We formulate disocclusion reliability learning as predicting teacher usefulness from test-time observable features. At the *scene* level the problem is data-limited: a simple visible-gap rule is already a strong policy and a learned scene-level ReliabilityNet only matches it (N=166 scenes: rule +0.87 dB vs learned +1.13 dB, both below oracle +1.70 dB). The key result is that moving to the *frame* level unlocks learning: with 2,898 frame-level samples a logistic frame ReliabilityNet reaches **0.87 accuracy, ROC-AUC 0.95, PR-AUC 0.96**, and +1.60 dB **mean invisible-PSNR gain** — **94% of the oracle’s +1.70 dB** — **clearly beating the visible-gap rule (+0.89 dB) while saving 41% of teacher calls**. A tunable risk-averse operating point drives the worst case toward zero while remaining net-positive. Descending further to the *patch* level, prediction becomes hard (ROC-AUC 0.72) and the gain over always-injecting nearly vanishes, identifying the frame as the sweet spot. Across granularities the governing mechanism is stable: the correlation between the geometry-expert-vs-baseline invisible gap and the actual teacher improvement is +0.98.

1 Introduction

A generative disocclusion prior (e.g. a diffusion backbone guided by a feed-forward geometry expert) can substantially improve novel-view quality in disoccluded regions, but it is expensive and *not uniformly beneficial*: on scenes where the feed-forward baseline already reconstructs the invisible region well, invoking the generative teacher hurts. This makes reliability prediction a first-class problem: we want to call the teacher only when, and ideally only where, it will help.

We formulate this as supervised reliability learning from test-time observable features (the geometry expert’s self-metrics, camera motion, disocclusion statistics), with labels given by whether the teacher actually reduces invisible-region error. Our study answers three questions: (i) is scene-level reliability learnable? (ii) does finer granularity help? (iii) is there a lower bound on useful granularity? We find that the frame level is the sweet spot: it provides enough samples for a learned predictor to beat a strong rule and approach the oracle, while exposing a tunable mean-vs-worst-case frontier.

Contributions.

1. We formulate disocclusion reliability as a learnable scheduling problem with purely test-time-observable features and teacher-usefulness labels.
2. We show a granularity effect: scene-level is data-limited (learned $\approx$ rule), frame-level unlocks learning (ROC-AUC 0.95, 94% of oracle, 41% compute saved), and patch-level is hard (ROC-AUC 0.72), so the frame is the sweet spot.
3. We expose a tunable mean-vs-worst-case operating frontier and a stable +0.98 governing mechanism reproduced across scales.

2 Related Work

Our predictor is a form of test-time routing / cascade: decide from cheap observable signals whether to invoke an expensive expert. This is related to model cascades and deferral, but instantiated for single-view 3D disocclusion, where the “expert” is a generative prior [7, 3, 4] over a feed-forward reconstructor [5, 1, 2, 6]. The governing signal we exploit — the invisible-region gap between the geometry expert and the generative baseline — is the same mechanism identified for selective test-time injection in the companion paper, here turned into a learned, deployable scheduler.

Table 1: Scene-level policies (N=166, LOOCV). Learned only slightly edges the rule; scene features saturate.

Policy	Mean Δ	Worst	In
Always teacher	+0.75	-11.72	166/
Rule $\text{vis_gap} > 0$	+0.87	-8.51	162/
Scene ReliabilityNet (LOOCV)	+1.13	-6.80	114/
Oracle	+1.70	0.00	98/

Table 2: Frame-level policies (2,898 frames, grouped-CV). The learned predictor beats the rule by a wide margin and reaches 94% of the oracle while saving 41% of teacher calls.

Policy	Mean Δ	Worst	ROC-AUC	Saved
Always teacher	+0.48	-23.09	—	100%
Rule $\text{vis_gap} > 0$	+0.89	-10.63	—	14%
Frame ReliabilityNet	+1.60	-2.61	0.949	41%
Oracle	+1.70	0.00	—	0%

3 Method

Problem. For each unit (scene / frame / patch) we predict a binary label: will the generative teacher reduce invisible-region MSE relative to the feed-forward baseline? Features are test-time observable: the geometry expert’s visible/invisible self-PSNR, the visible gap (expert visible – baseline visible), disocclusion fraction, and — at frame level — camera translation and rotation from the source view plus interaction terms.

Predictor. A logistic ReliabilityNet is trained under grouped leave-one-scene-out cross-validation so that no frames/patches from a test scene appear in training (no leakage). At test time a threshold on the predicted probability decides whether to invoke the teacher, tracing a mean-vs-worst-case operating frontier.

4 Experiments

4.1 Scene-level: the problem is data-limited (N=166)

The learned net’s margin over the rule grows monotonically with data (parity at N=92; +0.11 at N=129; +0.19 at N=149; +0.26 at N=166), indicating the scene level is data-limited rather than unlearnable.

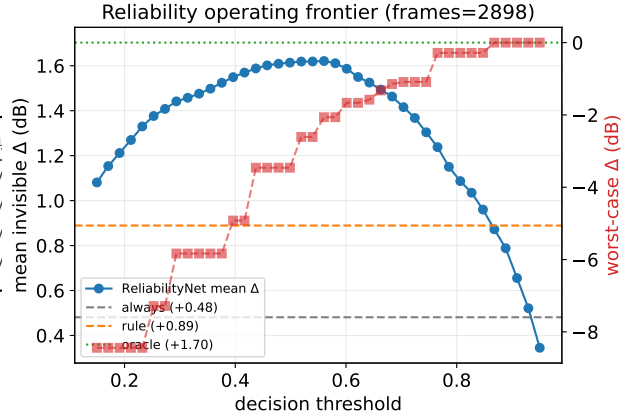


Figure 1: Mean-vs-worst-case operating frontier as the decision threshold varies. A fixed rule is a single point; the learned predictor spans a frontier, letting an operator choose maximum quality or near-zero-risk operation.

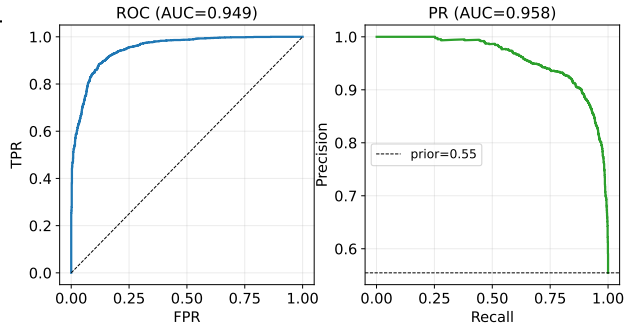


Figure 2: Frame-level ROC and precision-recall curves (grouped-CV). ROC-AUC 0.95, PR-AUC 0.96.

4.2 Frame-level: learning unlocks the gain (2,898 frames, 74 scenes)

The frame ReliabilityNet reaches accuracy 0.87, ROC-AUC 0.949, PR-AUC 0.958 (0.961 with camera-motion features). As the pool grows to 2,898 frames the newly added scenes have lower average teacher gain (always-inject drops to +0.48 dB), yet the learned predictor still recovers +1.60 dB — the harder the pool, the more selective scheduling matters. Figures 1, 2, 3 show the operating frontier, ROC/PR curves, and compute-vs-quality trade-off.

4.3 Patch-level: the granularity has a lower bound

Using 8×8 patch tiling with only test-time-observable patch features (geometry-expert-vs-baseline L1 disagreement and its variance, baseline gradient, disocclusion fraction, patch position, expert activity), patch

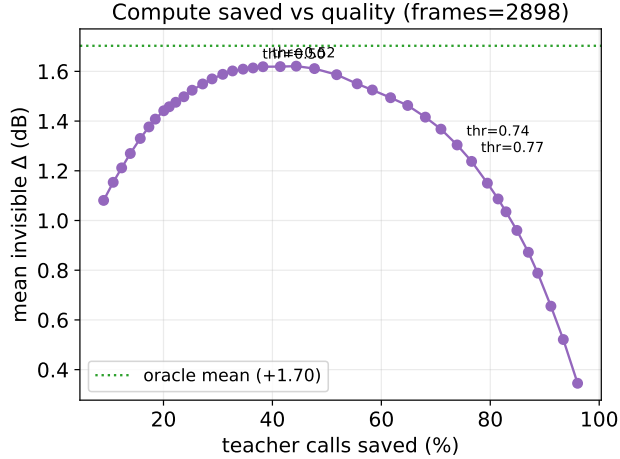


Figure 3: Compute saved vs quality retained: the predictor keeps most of the oracle gain while skipping a large fraction of teacher calls.

Table 3: Routing-policy comparison at matched budget (59% calls = 41% compute saved). Our learned scheduler beats random, always-call, and top-K observable ranking at the same cost, reaching 94% of the oracle.

Routing policy	Mean Δ (dB)	Note
Random @59%	+0.26	uninformed
Always call	+0.48	0% saved
Vis-gap top-K @59%	+1.08	strongest rule
FrameNet (ours) @59%	+1.60	learned
Oracle	+1.70	upper bound

reliability is substantially harder to predict (ROC-AUC 0.724 vs 0.95 at frame level, 13,831 patches / 21 scenes) and the learned patch gate barely improves over always-injecting (+3.20 vs +3.05 dB, far from the +4.15 dB oracle). On high-gain scenes 71% of disoccluded patches are already teacher-favorable, so the within-frame signal is weak. This is an honest negative boundary result: the frame is the sweet spot — coarse enough for a strong learnable signal, fine enough to capture per-view variation the scene level misses.

4.4 Comparison with Standard Routing Baselines

Reliability routing is an instance of expert deferral / model cascades. We compare against standard routing baselines at matched compute budget ($\sim 59\%$ teacher-call rate, i.e. 41% saved):

4.5 Stable mechanism across granularities

The correlation between the observable invisible gap ($\text{expert}_{\text{inv}} - \text{baseline}_{\text{inv}}$) and the actual teacher improvement is +0.982 (scene, $N=166$) and +0.978 (frame, 2,898). The mechanism governing when the generative prior helps is the same at both scales and highly predictable.

5 Discussion

The arc is: (i) disocclusion reliability is a real supervised problem; (ii) at the scene level it is data-limited and a visible-gap rule is a strong interpretable baseline; (iii) descending to the frame level provides the sample size that lets a learned predictor beat the rule and approach the oracle while exposing a tunable mean-vs-worst-case frontier; (iv) descending further to the patch level, prediction becomes hard and the gain over always-inject nearly vanishes. The frame is the sweet spot. The practical contribution is the frontier: an operator can pick maximum quality (+1.60 dB) or a conservative near-zero-risk operating point.

6 Limitations

The predictor uses observable proxy features rather than the true (unobservable) teacher gain; its ceiling is the oracle it approximates (94% reached). Patch/pixel-level selection would need richer per-pixel evidence (e.g. depth uncertainty) to pay off. Scaling the frame set further and integrating the predictor as a learned scheduler into the generative-injection and distillation pipelines are natural next steps.

7 Conclusion

We established disocclusion reliability as a learnable scheduling problem and delivered a working frame-level predictor that reaches ROC-AUC 0.95, recovers 94% of the oracle gain, and saves 41% of teacher calls under an honest grouped-CV protocol, with a stable +0.98 governing mechanism reproduced across scales and a clear granularity sweet spot at the frame level.

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